Packages in Java

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A Java package is a group of similar type of classes, interfaces and sub packages.

What is Package in Java?

The Java package groups related classes, interfaces and sub packages into one unit. It helps organize large programs, avoid name conflicts, control access, and make code reusable and modular.

There are two types of Java packages:

- Built-in Packages
- User-defined Packages

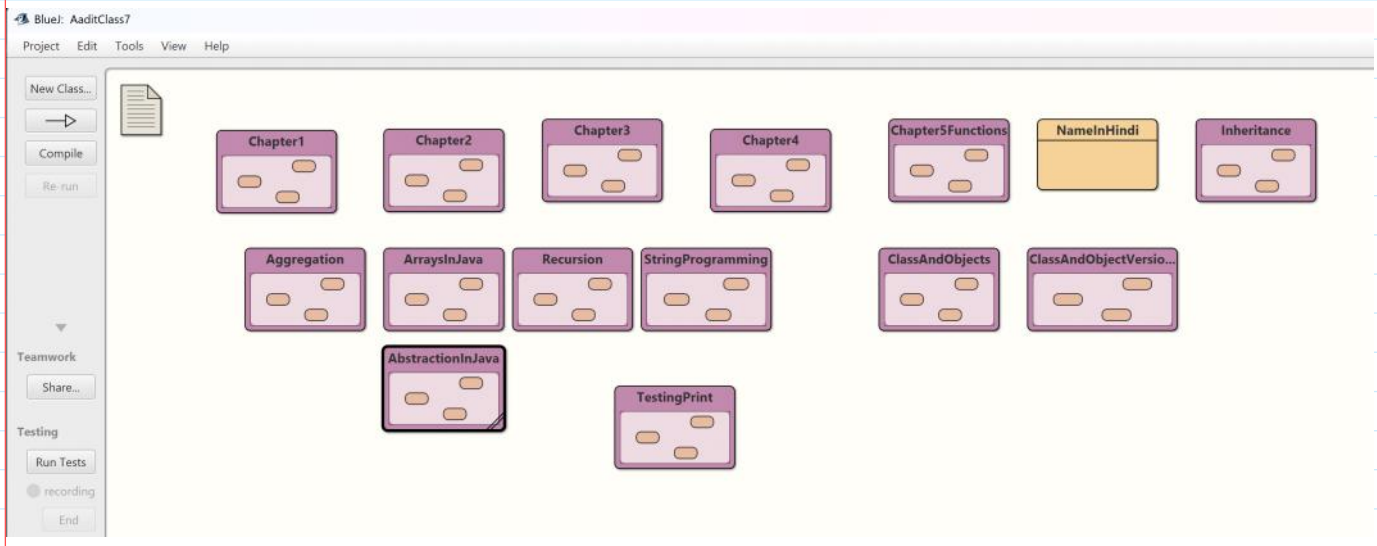
Java Built-in Packages

Built in packages are pre defined packages provided by Java that contain ready to use classes and interfaces. Some of the built in packages that are commonly used are:

- **lang**
- **util**
- **io**
- **javax.swing**
- **… many more, as we go we learn them**

Java User-Defined Packages

User defined packages are packages created by programmers to group their own classes and interface. They help to avoid name conflicts, support access control, and allow developers to structure large projects in a clean and modular way.




```
package AbstractionInJava.InterfacesInJava;
```

← In every program in our project(s), this must be the 1st statement.

```
/**
 * Write a description of class MyPrint here.
 *
 * @author (your name)
 * @version (a version number or a date)
```

```
 */
public class MyPrint extends AbstractPrinting
{
    @Override
    public void print(){
        System.out.print("\nThis is AbstractPrinting.\n");
    }
}
```

Skill:

Static import of the packages

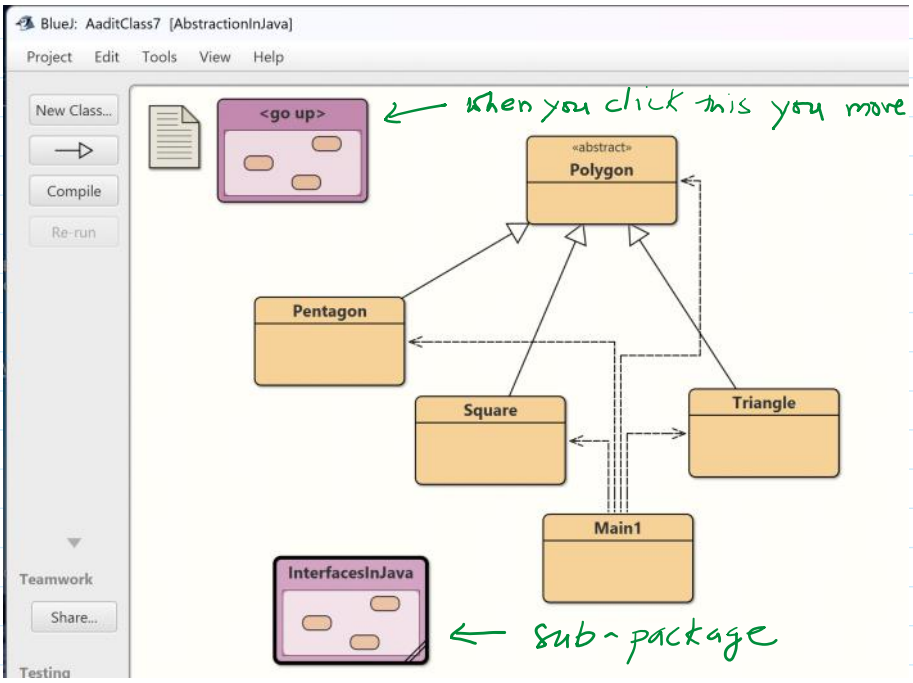
The static import feature was first introduced in Java version 5 and is still present in the latest version of Java. Static import allows only public static members (methods or fields) of a class to be used in the code without specifying the name of the class in which it is defined.

```
package AbstractionInJava.InterfacesInJava;
import static java.lang.System.*;
import static java.lang.Math.*;

/**
 * Write a description of class TestStaticImport here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class TestStaticImport
{
    public static void main(String[] args){
        out.print("\f");
        out.print("2^3 = " + pow(2, 3));
    }
}
```

Java Subpackage

A package inside a package is called a sub package. It should be created to categorize the package further.



← when you click this you move to the parent package.

← sub-package

```
package AbstractionInJava.InterfacesInJava;
```

super package sub-package.

```
/**
 * Write a description of class MainClass here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class MainClass
{
    public static void main(String[] args){
        System.out.print("\f");
        Printer p = new Printer();
        p.msg();

        MessagePrinter mp = new MessagePrinter();
        mp.msg();
        mp.printMsg();
    }
}
```

Importing classes from different package in your code:

use to import all the classes and interface present in the sub-package.

```
package TestingPrint;
import AbstractionInJava.InterfacesInJava.*;
import java.util.*;
```

super package sub package

```
/**
 * Write a description of class PrintingPattern here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
```

```
public class PrintingPattern
{
    /**
     * 1
     * 1 2
     * 1 2 3
     * 1 2 3 4
     * 1 2 3 4 5
     */

    public static void main(String[] args){
        P p = new P();
        p.print("\f");
        int i, n;
        Scanner sc = new Scanner(System.in);
        p.print("Enter number of lines: " );
        n = sc.nextInt();
        for(i = 1; i <= n; i++){
            for(int j = 1; j <= i; j++){
                p.print(j+" ");
            }
            p.println();
        }
    }
}
```